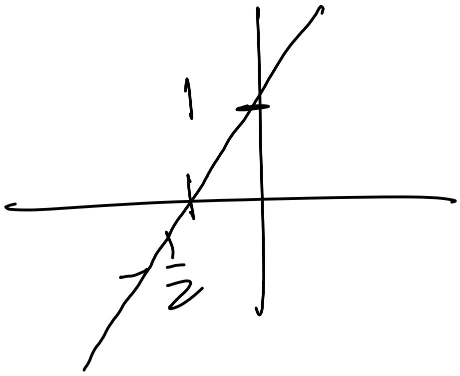
Unit 1.5 A library of functions p 23–35.

Note Title 2015/02/08

1. Linear functions

$y = ax + b$

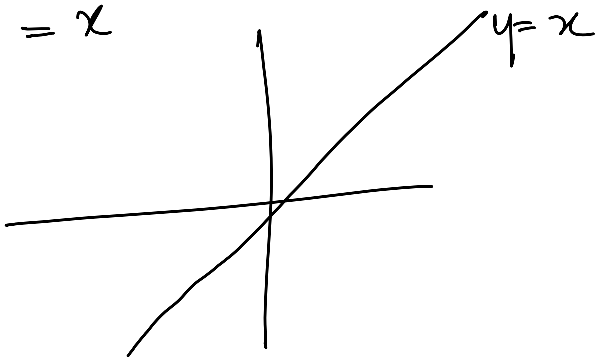


$y = 2x + 1$

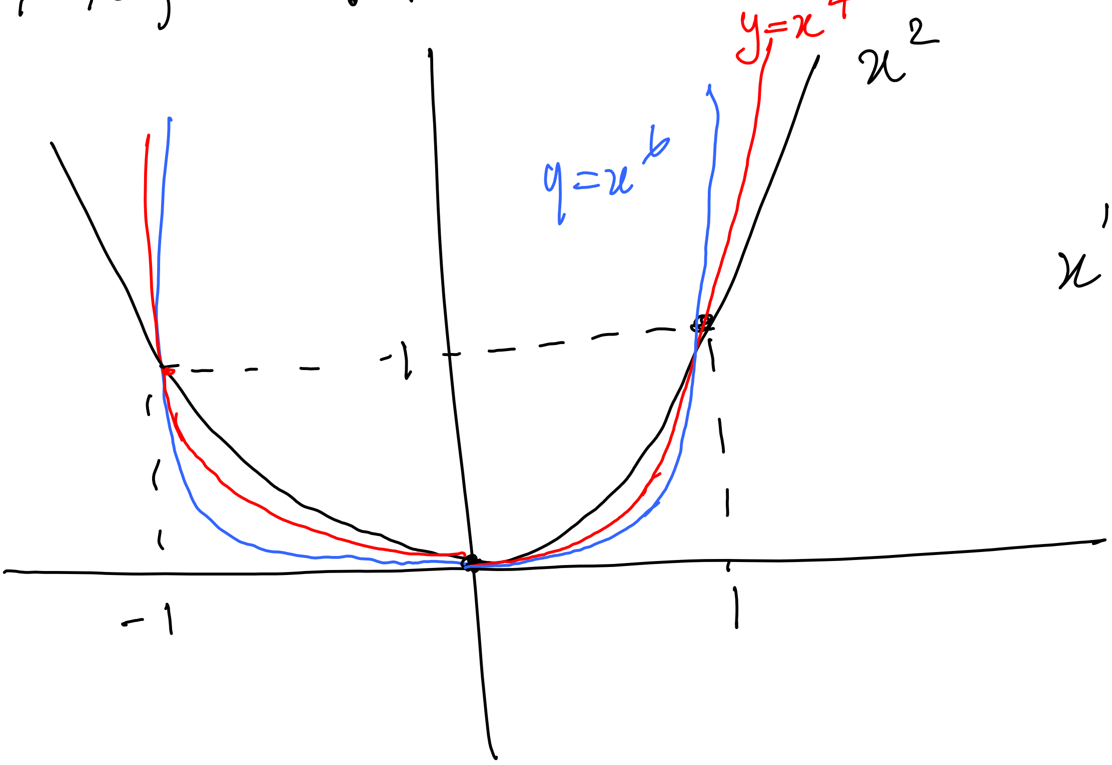
2. Power functions : $y = x^3$

$y = x^n$

- $n = 1$: $f(x) = x$

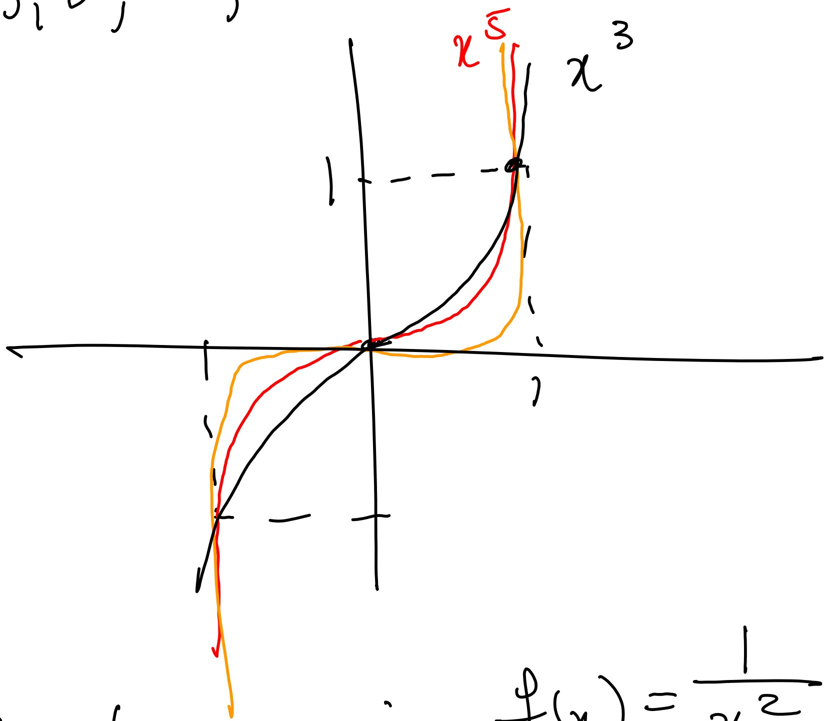


- $n = 2, 4, 6, \ldots$: $f(x) = x^2$ etc.



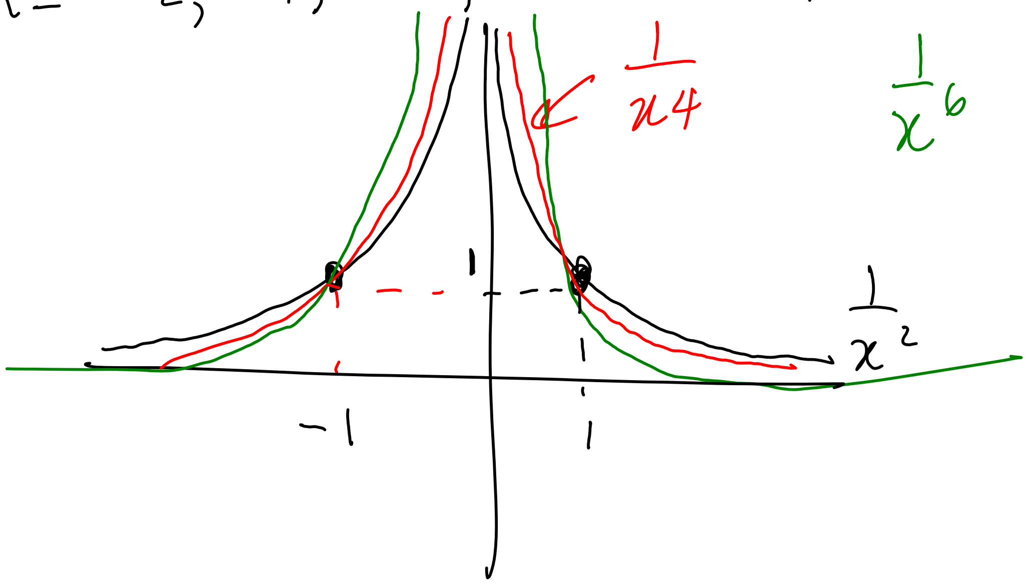
x^{100} ?

• $n = 3, 5, 7, \dots$: $f(x) = x^3$ etc.



odd. → $y = x^{101}$?

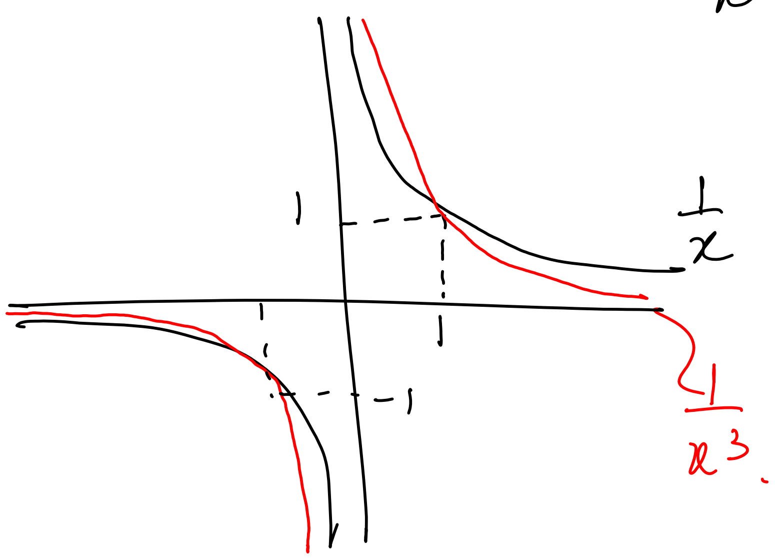
$n = -2, -4, -6, \dots$: $f(x) = \frac{1}{x^2}$ etc.



$\frac{1}{x^4}$

$\frac{1}{x^{100}}$?

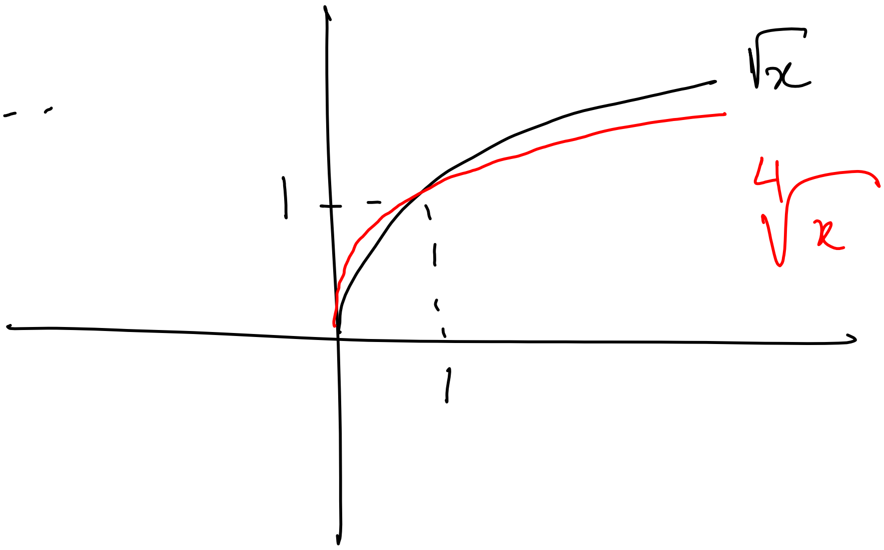
$n = -1, -3, -5, \dots$: $f(x) = \frac{1}{x^3}$ etc.



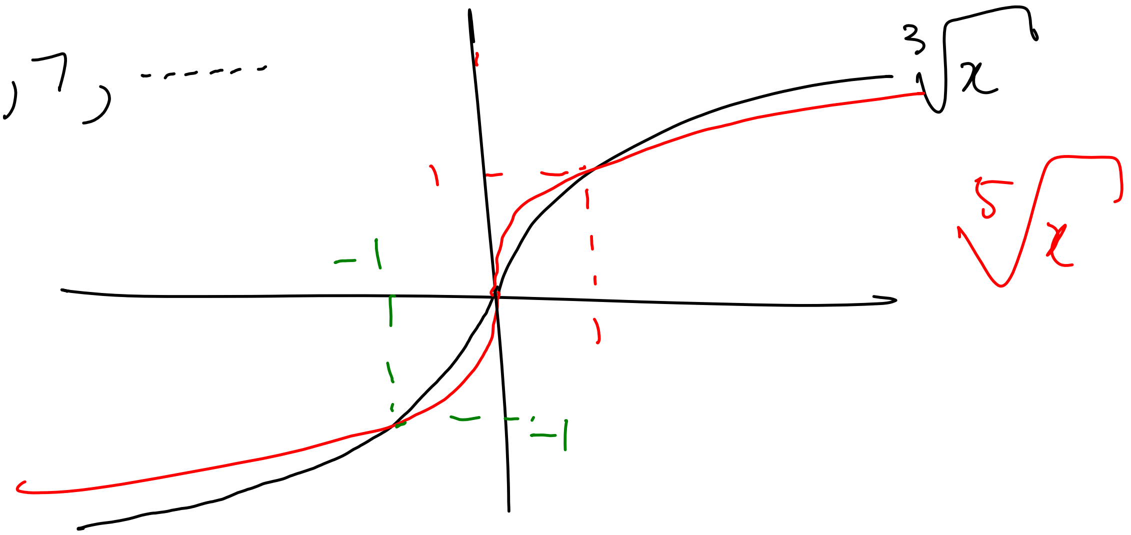
• $f(x) = \sqrt[n]{x}$

$n = 2, 4, 6, \ldots$

$\sqrt[100]{x}$



• $n = 3, 5, 7, \ldots$



3. Polynomials

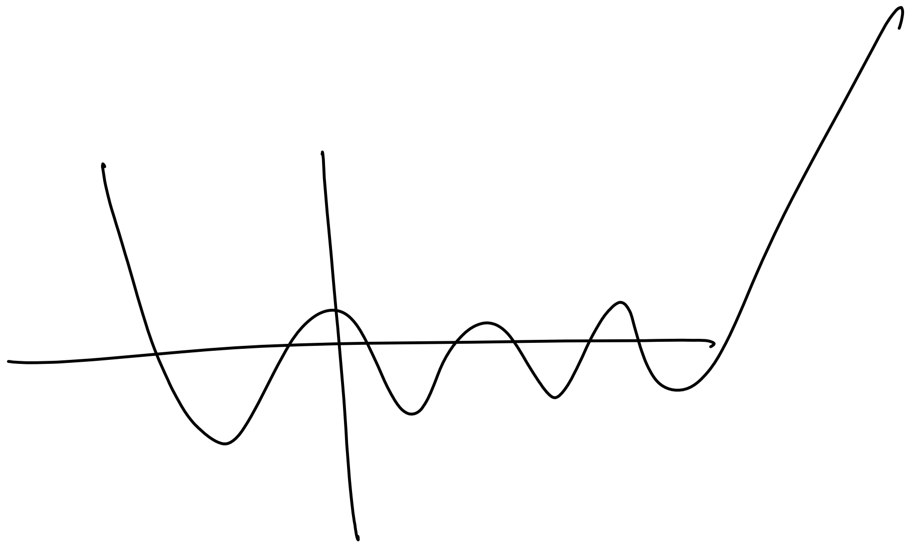
$a_n \neq 0$

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

Eg $f(x) = x^4 - 3x^2 + x - 1.$

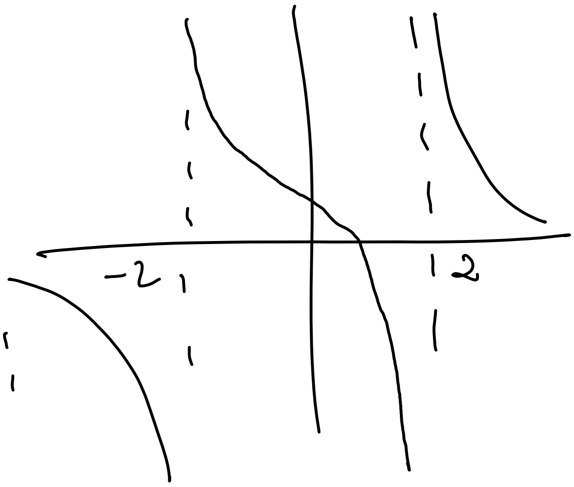
Polynomial of degree n:

- has at most $n-1$ turning points
- has at most n x-intercepts.



4. Rational functions

$$f(x) = \frac{P(x)}{Q(x)} \begin{matrix} \text{— polynomial} \\ \text{— polynomial} \end{matrix}$$



- Domain $\{x \mid Q(x) \neq 0\}$
- Known for having asymptotes!

Eg. $f(x) = \dfrac{x-1}{x^2-4}$

Algebraic functions

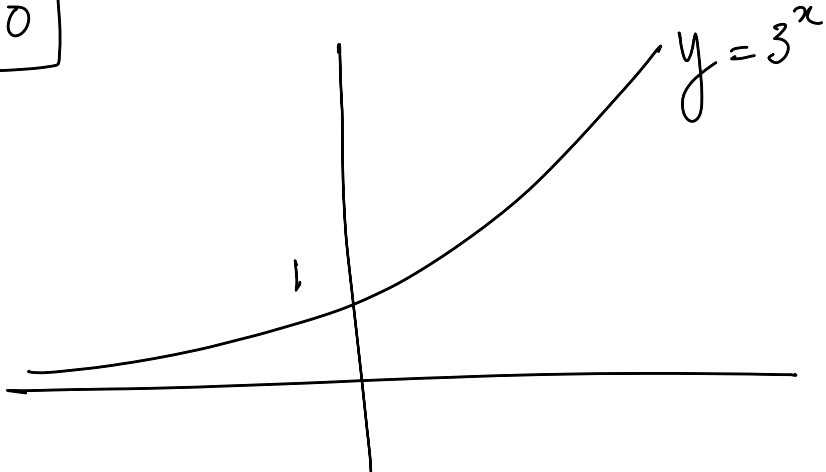
Polynomials on which operations have been performed

eg $\dfrac{\sqrt{x^2+1}}{x-1}$

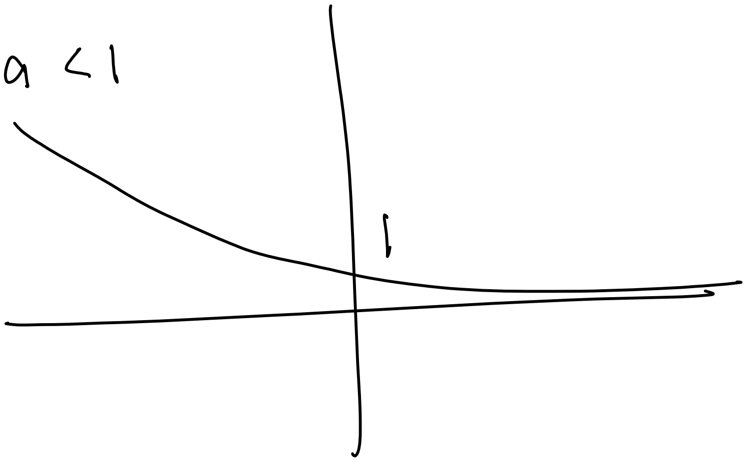
5. Exponential functions

$f(x) = a^x$, $\boxed{a>0}$

$a > 1$



$0 < a < 1$

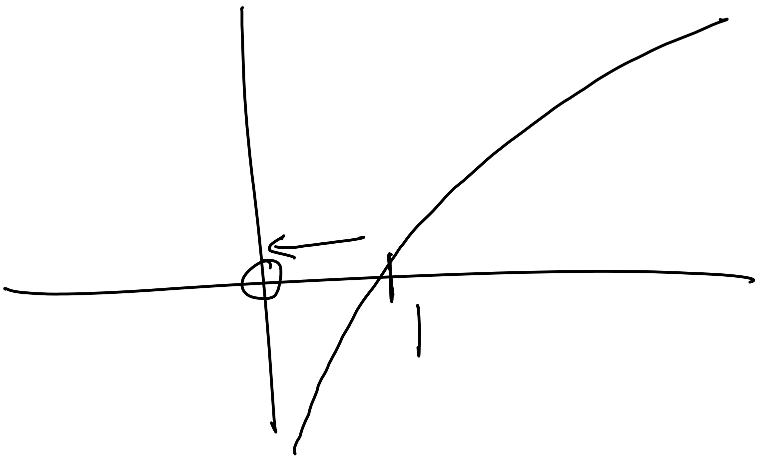


$y = \left(\frac{1}{2}\right)^x$

$= 2^{-x}$

6. Logarithmic functions.

$f(x) = \log_a x \quad , a > 0$



$\log\left(\frac{1}{100}\right) = -2$

$\log 10^6 =$

7. Trigonometric functions.

$y = C + A \sin B x$

Period: $\frac{2\pi}{|B|}$

Amplitude $|A|$

y-intercept: C